

Achieving High Safety for Lithium-Ion Batteries by Optimizing Electron and Phonon Transport

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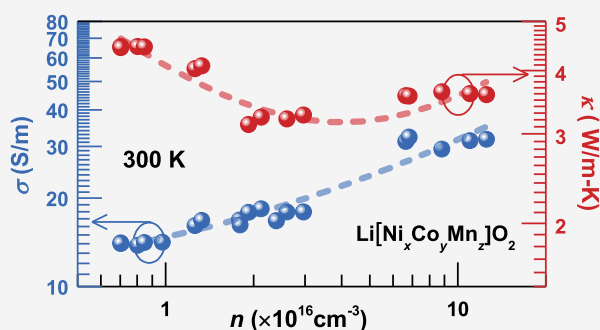


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Supporting Information

ABSTRACT: For lithium-ion batteries, frequent safety issues remain a huge challenge for safe applications. Herein, we focus on the coupled effect of electronic transport and phonon scattering on the thermal safety of $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ cathode materials. It is revealed that the electric conductivity of $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ doubles to 33 S/m as Ni content increases from 1/3 to 0.8. However, the closure in the band gap leads to a strong compensation of phonon scattering due to the thermal motion of lattice atoms. The collision between electrons and phonons damages the periodicity of the potential field, leading to a much stronger phonon scattering and ultimately a 22% decrease in thermal conductivity. We demonstrate that an improvement in electronic conduction can result in a stronger lattice vibration that deteriorates the thermal safety of the battery. Understanding the electronic/thermal transport mechanism provides new guidance for solving the basic issue of heat generation, which is critical for designing Li-ion batteries with high safety.



Lithium-ion batteries (LIBs) have important applications in portable electronic devices and electric vehicles (EVs) because of their high energy density and long service life.^{1–3} However, the safety issue associated with LIBs remains a huge challenge for practical applications, which is attributed to severe thermal runaway (TR) of the battery.^{4,5} After decades of development, there are many effective strategies to improve the thermal safety of EV batteries.^{6–8} Typical strategies focus on materials, cells, and modules,^{9,10} including reduction of oxygen evolution in cathode materials, inhibition of lithium precipitation, functionalization of separators to avoid internal short circuit, and application of insulation to prevent the spreading of TR.^{11–14} However, the TR of the battery is mainly rooted in the local heating induced by both exothermic electrochemical reactions and the Joule heat generated from the charging–discharging process, including chemical reduction/oxidation of the anode/cathode and thermal decomposition of the electrode and electrolyte.^{15,16} Therefore, enhancing the internal thermal transport performance of LIBs during cycling is of critical importance for battery safety.

Increasing the thermal conductivity (κ), which involves both electronic (κ_E) and lattice (κ_L) contributions of electrode materials, is an effective strategy to improve the internal thermal transport performance of LIBs. The electronic

component of thermal conductivity is determined by the Lorenz factor (L) and electrical conductivity (σ) through the Wiedemann–Franz law ($\kappa_E = L\sigma T$, where T is temperature),^{17,18} and thus improving σ by increasing the solid carrier concentration can effectively enhance the κ_E . But for many non-metallic materials with low electrical conductivity, κ_E makes only a trivial contribution to the total κ , and κ_L dominates the total κ . According to the Boltzmann transport theory of phonons, κ_L is determined by the specific heat (C_V), phonon group velocity (v_g), and phonon relaxation time (τ), which are all strongly linked to the dispersion of phonons, including acoustic phonons and optical phonons. Compared to optical phonons,^{17,19} acoustic phonons are the main contributors to κ_L because they have much higher velocities, and thus reducing the scattering rate of acoustic phonons is the main route to intensify the lattice thermal conductivity.^{20–22} However, due to the interaction between electrons and lattice vibration, when the lattice atomic vibration is offset (scattering

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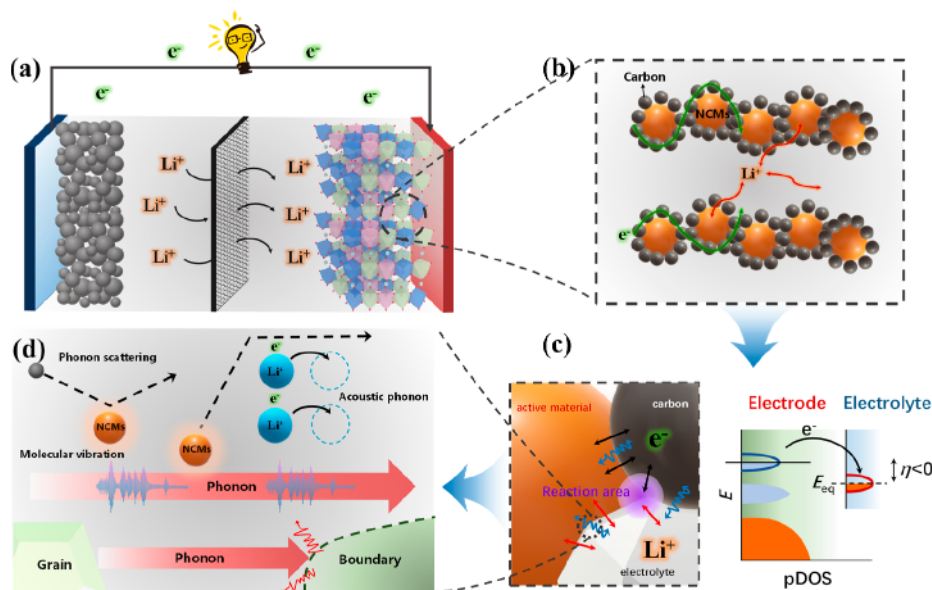


Figure 1. (a) Schematic diagram of the working mechanism of a lithium-ion battery. (b) Diagram of the ion and electron transport mechanism in a positive electrode. (c) With ion intercalating/deintercalating, the electron population of the host structure changed, resulting in changes in both the electronic conductivity of the electrode and the thermodynamic electrochemical driving force of the reaction ($\eta = E_F - E_{eq}$, where E_F is the Fermi level of the electrode and E_{eq} is the Fermi level of the reactant in solution). (d) Phonon transport mechanism of a cathode electrode.

takes place), there is an exchange of momentum and energy between electrons and lattice vibration, resulting in coupling between electrons and phonons.^{21,23} In addition, the micro-mechanisms of the electrolyte/electrode interactions, including ion insertion, are complex because they are closely dependent on the properties of the cathode materials, the redox couple, the intercalated ions, and the electrolyte. The charge/discharge processes include the desolvation of Li ions in the electrolyte (Figure 1a), the interfacial transport of the ions from the electrolyte into the host lattice, and the transfer of electrons from the host lattice. The ions and electrons become localized during these processes (Figure 1b), so the migration of the Li ion occurs via the ambipolar hopping of an ion–electron pair²⁴ (Figure 1c), with phonons transporting through the electrode²⁵ (Figure 1d), which demonstrates that the electron/phonon transfer rate determines the speed of redox reactions in the cell.^{26–29} Therefore, understanding the role of electrical/thermal transport and the electron–phonon coupling mechanism in electrode materials is urgent and essential toward safe high-energy battery design. The layered $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ (NCM) ternary cathode materials have attracted intensive research interest for high-performance LIBs due to their high energy density.^{30–32} However, the application of NCM ternary metal oxides, especially Ni-rich ones, seriously suffers from TR and safety issues.^{33,34}

In this work, we focus on the electronic and thermal performance of different $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ cathode materials to explore the coupling effect of electronic transport and phonon scattering on NCM cathodes. We discover that reducing the band gap and increasing the Hall mobility by increasing the Ni content in NCM can accelerate the electronic conduction to improve the electrical transport performance but significantly affects the thermal motion of lattice atoms and generates strong phonon scattering to decrease the thermal conductivity. In other words, there is a trade-off between electronic conduction and thermal transport properties for

cathode materials. Understanding such relationships offers new avenues for designing highly safe LIBs.

We prepared four conventional $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ ternary cathode materials: NCM111, NCM523, NCM622, and NCM811 ($x:y:z = 1:1:1, 5:2:3, 6:2:2, 8:1:1$, and $x + y + z = 1$). To obtain dense pellet samples for testing the thermal properties of NCMs, all powders were sintered at 1000 °C in an oxygen atmosphere after cold pressing with high pressure to reduce the cationic mixing and the impurity phases for $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$, which can be confirmed by Figure S5 and Table S1. Meanwhile, in order to fit the requirements for electrical property measurement, the thick electrodes ($\sim 300 \mu\text{m}$) without an Al collector were used to test the conductivity (σ) and carrier concentration (n). Details on the synthesis, calculations, characterizations, and electrochemical measurements are given in the Supporting Information (Figures S1–S5).

Traditionally, the electronic/ionic conduction networks are constructed of active material, carbon, and electrolyte, wherein the ions and electrons meet together at the triple-phase interfaces (Figure 1c). Commonly, the properties of ion insertion can be explained by point defects in the cathode materials. Point defects can significantly alter the electronic structure by altering the Fermi level or creating new bands, thus affecting the electrical transport properties of the cathode materials. Figure 2a shows the Ni content-dependent Hall carrier concentration (n_H) and carrier mobility (μ_H) for the synthesized $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$. n increases by an order of magnitude from 0.8×10^{16} to $9 \times 10^{16} \text{ cm}^{-3}$ when the Ni content increases from 1/3 to 0.8, which means better electronic transport properties for NCM with higher Ni content. Almost no change is observed in the carrier concentration within the measured temperature range, indicating that a single type of carrier contributes to the conduction in NCMs (Figure S6a). Moreover, μ_H is also found to increase with increasing Ni content (Figure 2a and Figure

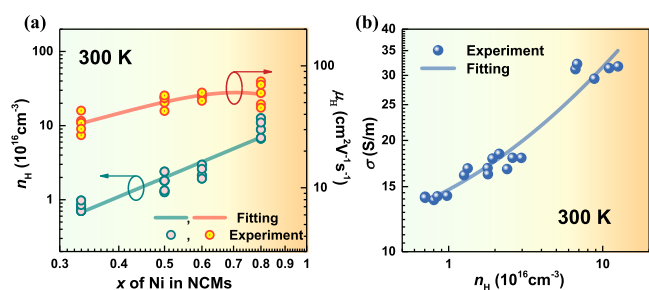


Figure 2. (a) Composition-dependent carrier concentration and Hall mobility and (b) carrier concentration-dependent electrical conductivity for $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ cathodes at room temperature.

S7), indicating that increasing the Ni content is indeed beneficial for enhancing the electronic performance. The thermal activation effect of polycrystalline $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ migration comes from the amorphous structure of the grain boundary, which changes the charge distribution among the interfacial layer and therefore results in a change of the barrier height. The large grains would be helpful to realize a charge-scattering mechanism closer to that of acoustic scattering and thus to achieve a high mobility. Therefore, single-crystalline $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ shows better performance compared with polycrystalline $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$. n depends on the Fermi level and the density-of-states (DOS) of the materials. It is found that n_H is nearly independent of temperature (Figure S6a), suggesting a rigid band behavior for NCMs. Importantly, all the $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ cathodes show a dominant charge scattering by acoustic phonons, as evidenced by the temperature-dependent μ_H via $\mu \sim T^{-1.5}$ (Figure S6b), which demonstrates that phonon scattering accelerates electron transport and improves μ_H . As a result, the electrical conductivity of $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ cathodes doubles to 33 S/m when the Ni content increases from 1/3 to 0.8, due to the enhancement in both n_H and μ_H (Figure 2b and Figure S8). Near room temperature, the band gap of the NCMs is large and the carrier transition is limited, so the change with temperature is not obvious. As the temperature increases, the lattice vibration intensifies, resulting in enhanced phonon scattering; therefore, σ will decrease as the temperature increases. The electrochemical reactions in the battery consist of various processes with coupled contributions from both ions and electrons. The electronic conduction also plays a significant role in the rate capability of batteries.³⁵ Therefore, increasing the Ni content in $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ cathodes can accelerate electronic conduction and thus improve the rate capability. Detailed information about the calculation of electrical properties is given in the Supporting Information.

To further confirm the effect of Ni content on the electronic structure of NCM, DOS calculations for the four NCM compositions were carried out, as shown in Figure 3a and c–f. Detailed partial calculations of the O 2p orbital, the TM 3d orbital, and the Li s orbital are shown in Figure S9. Obviously, in all NCMs, there is little contribution from the Li s orbital in the energy range of -5 to 3 eV in the total DOS. However, when the Ni content increases, the Fermi energy shifts toward larger values, which indicates that the n_H of NCMs significantly increases, because more valence electrons fill in the 3d orbital of Ni than Mn and Co. The calculation results are consistent with literature reports.^{30,35–38} Furthermore, the experimental composition-dependent band gap measured through optical absorption reasonably agrees with the calculated ones and the

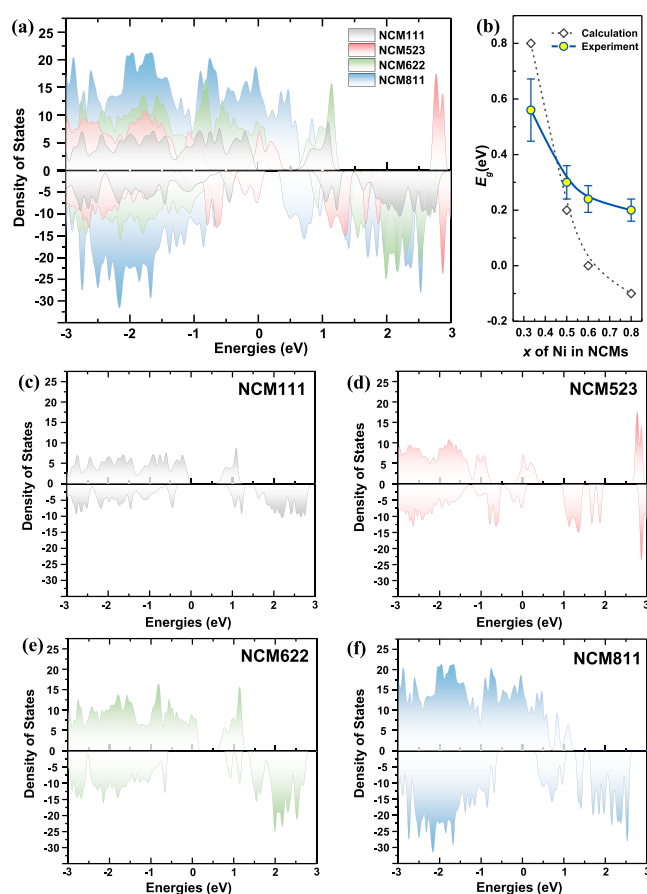


Figure 3. (a) Calculated band structures for NCMs. (b) Composition-dependent band gap. Individual band structures of (c) NCM 111, (d) NCM 523, (e) NCM 622, and (f) NCM 811.

literature results^{36,37} (Figure S10). Specifically, as shown in Figure 3b, a higher Ni content decreases the band gap from NCM333 (0.6 eV) to NCM811 (0.2 eV), resulting in the metallic properties of the NCMs. This further explains why increasing Ni content can effectively improve the electrical properties of NCM cathodes (Figure 2b and Figure S8).

Although narrow band gap and high mobility can enhance the electronic conduction of electrode materials,^{39,40} they also significantly affect the thermal motion of lattice atoms. The collision between electrons and phonons results in the periodic destruction of the potential field,^{23,41} which might cause much stronger lattice vibration and scattering to decrease the thermal conductivity, thus deteriorating the thermal transport properties of NCM cathodes. To explore the thermal transport properties, room-temperature sound velocities of NCM compounds are summarized in Table S1. Due to a strong anharmonicity (a large Grüneisen parameter of ~ 1.7 , Table S1), pristine NCM has low phonon relaxation times, implying a low thermal conductivity. As the Ni content increases, the longitudinal (v_l), transverse (v_t), and mean sound velocity (v_s) of NCM compounds decrease smoothly, which indicates that increasing the Ni content can reduce the thermal conductivity. Figure S11a displays the composition-dependent thermal conductivity for $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ at room temperature. Obviously, when the Ni content is increased in $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$, the thermal conductivity at room temperature expectedly decreases by 22%, from 4.5 W/m-K in NCM111 to 3.5 W/m-K of NCM811, implying that the

thermal transport performance becomes worse with increasing x .

To further verify the effect of electrons on the thermal transport properties in electrode materials, we theoretically calculated the thermal conductivity of $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ compounds through the Debye temperature.⁴² The results at room temperature are summarized in Tables S1 and S2. The numbers of atoms of the complex crystal structure induce the shrinking of the first Brillouin zone, which folds most of the high-frequency vibrational modes back as optical ones. Meanwhile, the velocities near zone boundaries significantly decrease, resulting in only a small portion of phonons being acoustic ones. However, the optical phonons' contributions to L are negligible due to their zero-like group velocities, leading to an actual reduction in specific heat of acoustic phonons at low L (C_p , Figure 4a and Figure S12).^{22,43,44} As shown in

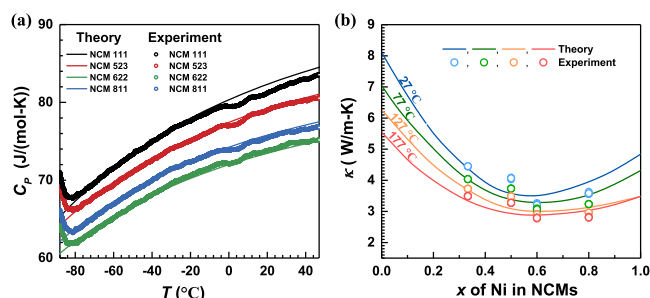


Figure 4. (a) Temperature-dependent specific heat and (b) composition-dependent thermal conductivity for $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ alloys, with a comparison to model predictions.

Figure 4a, the theoretical C_p curves fit well with the measured ones, which indicates that the thermal conductivity conforms to Debye's law.⁴² Based on the calculated Debye temperature (Table S1) and the Callaway model with the phonon dispersion considering the Born and von Karman periodic boundary conditions^{45,46} (Table S1), the theoretical κ values for $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ are obtained, as shown in Figure 4b. The relative coherence between the experimental and theoretical values in both C_p and κ proves the accuracy of the experimental values of κ . It is worth noting that both cross-plane and in-plane κ values (in the extrinsic conduction region) are almost proportional to $1/T$, indicative of the dominance of phonon–phonon Umklapp scattering in these materials (Figure S13). In addition, the κ curves at different temperatures show the same trend, demonstrating that this phonon scattering mechanism is suitable for $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ cathodes during the cycling process under variable operating temperatures. Interestingly, it can be seen from Figure 4b that the κ of $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ begins to increase at $x > 0.6$, due to the bipolar effect caused by the closure of the band gap. However, the effect of closing the band gap, which results in a strong bipolar conduction and bipolar thermal conductivity, might be critical. In addition, the measured κ values of LiCoO_2 and LiNiO_2 are found to be higher than the theoretical ones (Figure S11a), because the theoretical values include the effect of Mn^{4+} on the structural complexity of the crystal.

To determine the evolution of the thermal transport properties of different $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ compositions, Raman spectra were measured to analyze the molecular vibrations, as shown in Figure 5 and Table S2. $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ compounds show the spectroscopic D_{3d} symmetry, in

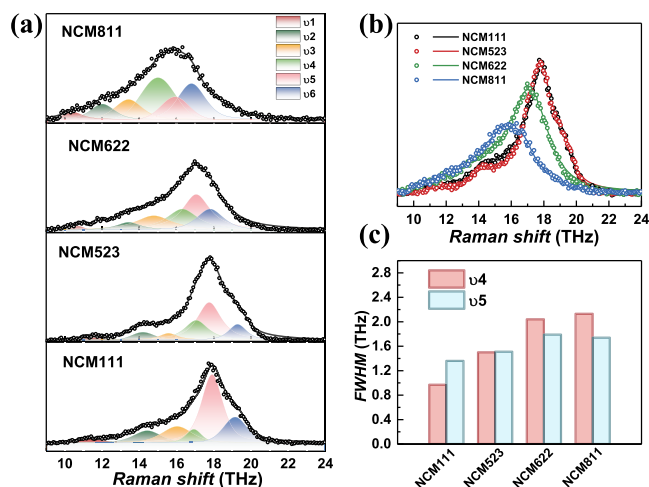


Figure 5. (a, b) Raman spectra with Lorentzian deconvolutions and (c) corresponding Raman peak broadening of the first transverse optical modes for $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ at room temperature.

which the vibration modes associated with each transition metal ion (M) are decomposed as follows:^{47–49}

$$\Gamma = 2A_{2u} + 2E_{2u} + A_{1g} + E_g \quad (1)$$

A_{1g} and E_g are Raman active modes that originate from the M–O symmetrical stretching and O–M–O bending vibrations, respectively. Since the compound has three M ions, we expect that the overlapped $3A_{1g}$ modes and $3E_g$ modes give rise to the two broad A_{1g} and E_g structures in Figure 5. For NCM111, the E_g bands are centered at 14.0, 14.5, and 15.3 THz, and the A_{1g} bands are centered at 16.4, 17.7, and 18.7 THz, for M = Ni, Co, and Mn, respectively (Figure 5a), which matches well with the corresponding bands of LiNiO_2 , LiCoO_2 , and LiMn_2O_4 , respectively. It is worth noting that these peaks shift consistently toward lower frequencies (Figure 5b), suggesting a reduction in ν_s with increasing Ni content (Table S1). Meanwhile, broadening in the full width at half-maximum for ν_5 offers important evidence of broadened phonon dispersion for a strong phonon scattering (Figure 5c), which also confirms the larger Ni cationic disorder in Ni-rich compositions, in agreement with the structural analysis. These results further demonstrate that the thermal conductivity is reduced by increasing the Ni content to improve the electrical properties.

A series of ~4-Ah typical pouch cells were prepared to verify the cycling stability (Figure 6a,b). The initial charge/discharge capacities of the cells can reach 4 Ah at a rate of 1/3C due to excellent electronic conduction, with good capacity retention of around 97% after 400 cycles. This helps us to exclude the influence of electrochemical properties and to better discuss the thermal properties. The detailed cycling performance of the pouch cells is shown in Figures S13 and S14. All the above-mentioned pouch cells were evaluated by using accelerating rate calorimetry (ARC) to clearly understand the effects of electron and phonon transport on the TR mechanism, as shown in Figure 6. T_1 is defined as the onset temperature of self-heating, T_2 is the TR trigger temperature, and T_3 is the maximum temperature during TR, which are key parameters to evaluate the destructiveness of TR.⁵⁰ To achieve high-safety LIBs, it is critical to increase T_1 , maximize T_2 , and minimize T_3 . With high enough content of Ni in cathode materials, although

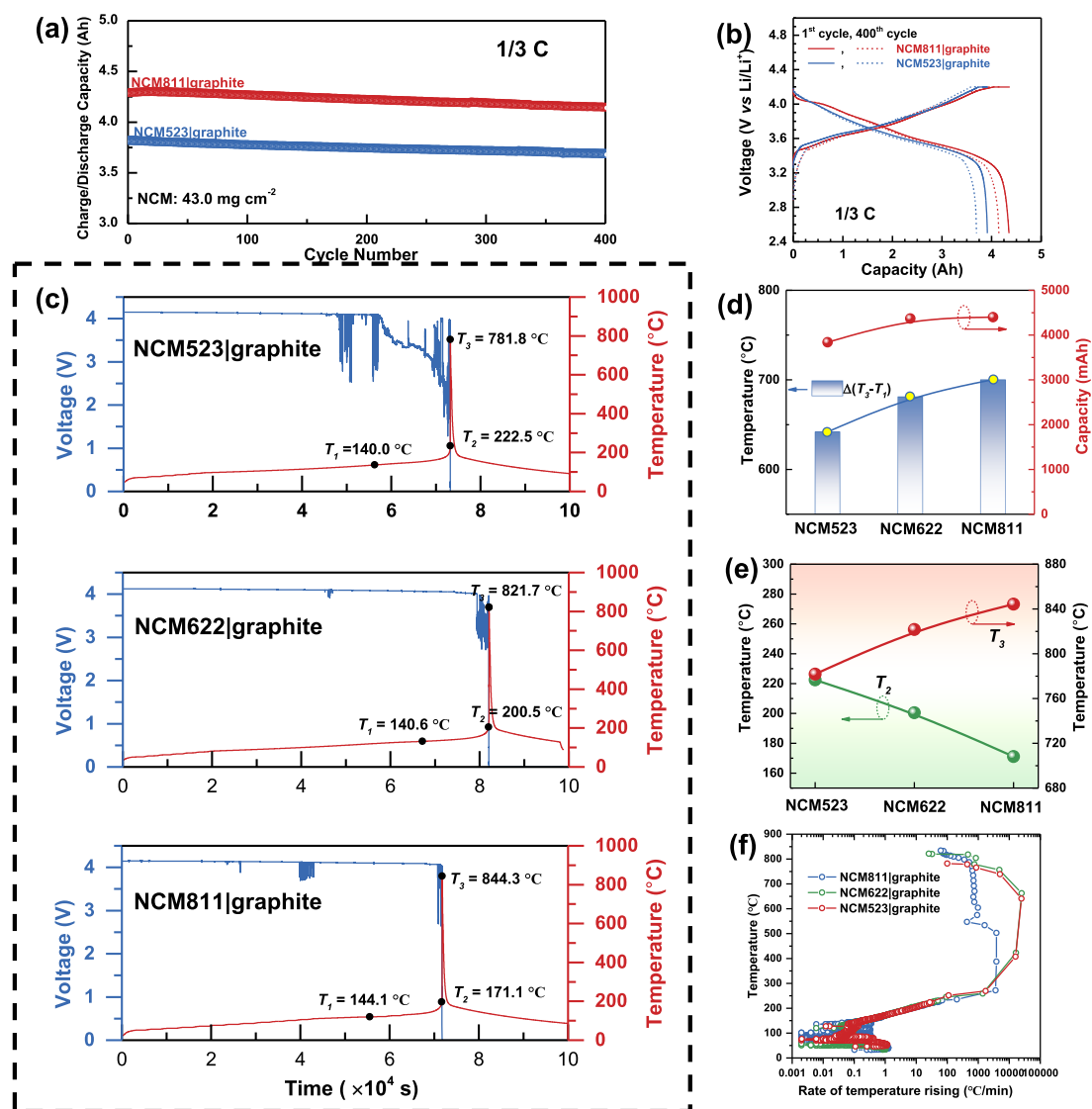


Figure 6. (a) Charge–discharge curves and (b) cycling stability of typical NCM523 and NCM811 cells cycled during 2.5–4.2 V. Intrinsic safety features of charged practical Ah-level pouch-type NCM|graphite cells under accelerating rate calorimetry test: (c) thermal runaway temperature plots, (d) composition dependence of the capacity and self-heating temperature difference ($\Delta T_3 - T_1$), (e) thermal runaway features, and (f) self-heating rate.

the κ has a slight rebound after the decline, the metallic properties and electrical conductivity are enhanced (Figure 2). However, the oxygen release reaction of the cathode materials is severe. The ARC test shows that T_2 of the cell decreases from ~ 223 to ~ 171 °C, while T_3 increases from 782 to 844 °C, implying that a higher Ni content significantly deteriorates the thermal stability of the cell. Interestingly, the ARC test results are almost consistent with the thermal transport properties of different NCMs. Although the thermal conductivity of NCM811 is slightly higher than that of NCM622, the ARC test result of the NCM811|graphite cell is inferior to that of NCM622|graphite, which is attributed to the higher reactivity of NCM811.^{7,51} Therefore, increasing the thermal conductivity of the electrode materials can effectively improve the thermal stability of the cell due to reduced heat generation and maintain the high capacity attributed to the high-speed migration of electrons.

We discover that increasing the carrier concentration and mobility by increasing the Ni content in NCM can effectively

increase the electrical conductivity, whereas the reduction of the band gap and enhancement of mobility can significantly affect the thermal motion of lattice atoms. Collisions between electrons and phonons result in the periodic destruction of the potential field and eventually cause much stronger lattice vibration and scattering, which decreases the thermal conductivity and deteriorates the thermal transport properties of NCM materials. According to the ARC test, with increasing Ni content in active cathode materials, the onset temperature of thermal runaway trigger of the cell decreases from 223 to 171 °C, while the maximum temperature during the thermal runaway increases from 782 to 844 °C, implying that a high Ni content obviously damages the thermal stability of the battery due to the increase of phonon scattering induced by high-speed migration of electrons. Therefore, selecting appropriate regulation methods based on the electron/phonon scattering mechanism is crucial for improving the heat dissipation, safety, and stability of lithium-ion batteries. More importantly, we demonstrate that optimizing the thermal safety of the battery

by improving the thermal transport properties of electrode materials sheds new light on the design principles of cathode and even anode materials for high-safety rechargeable batteries.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsenergylett.3c01787>.

Materials preparation section, additional information about characterization and measurements of $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ cathodes, information about cycling experiments and thermal runaway testing methods of NCM cells, electronic and structure calculations, and thermal conductivity modeling of $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ (PDF)

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Notes

The authors declare no competing financial interest.

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